

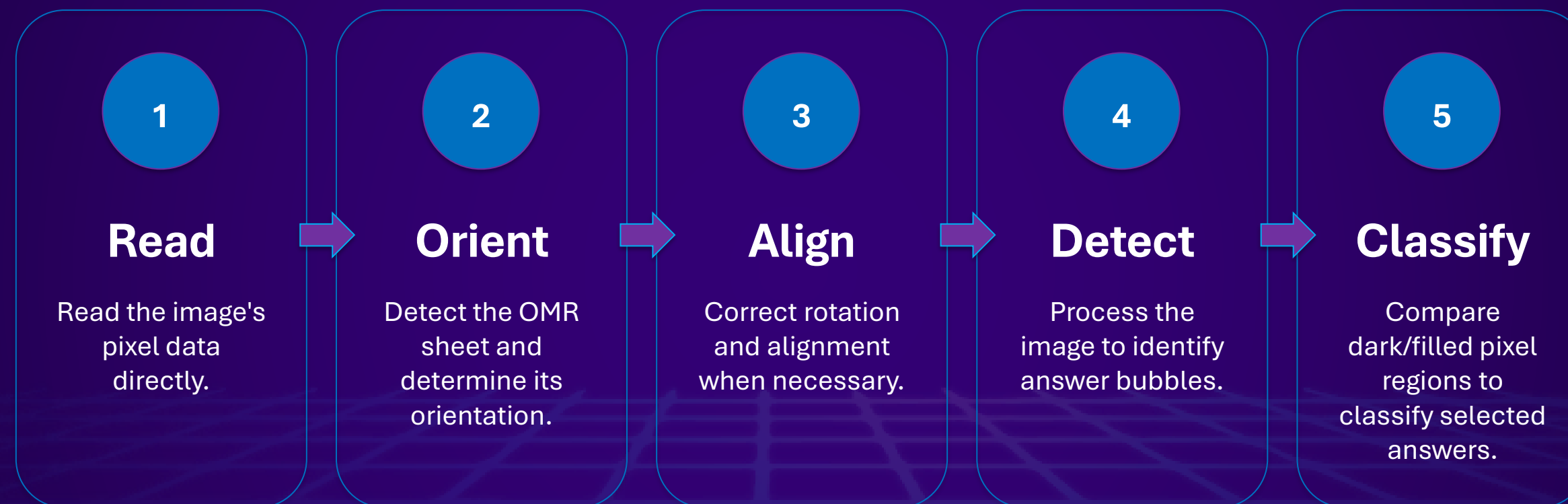
Native Optical Mark Recognition

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Proposed Solution

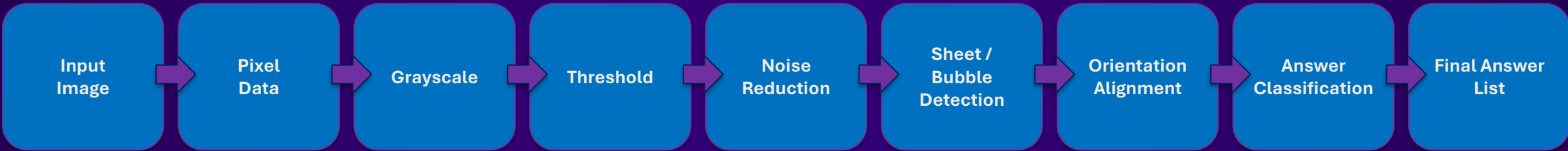
A direct pixel-processing workflow



Final result: display the detected answer list.

Image Processing & OMR Pipeline

From raw image to the final answer list



CUSTOM IMAGE PROCESSING

All core image-processing operations are implemented using custom routines.

Pixel data → Grayscale → Threshold → Noise reduction → Detection → Alignment → Classification

Expected Output & Development Plan

A sample result and the implementation roadmap

EXPECTED OUTPUT

Question	Answer
1	B
2	D
3	A
4	C
5	B
6	Unanswered

DEVELOPMENT PLAN

- 1 Image reading & pixel processing
- 2 Grayscale, thresholding & noise reduction
- 3 Sheet detection & orientation correction
- 4 Bubble detection & answer classification
- 5 Testing, optimization & final interface

No OpenCV • No Tesseract • Custom routines